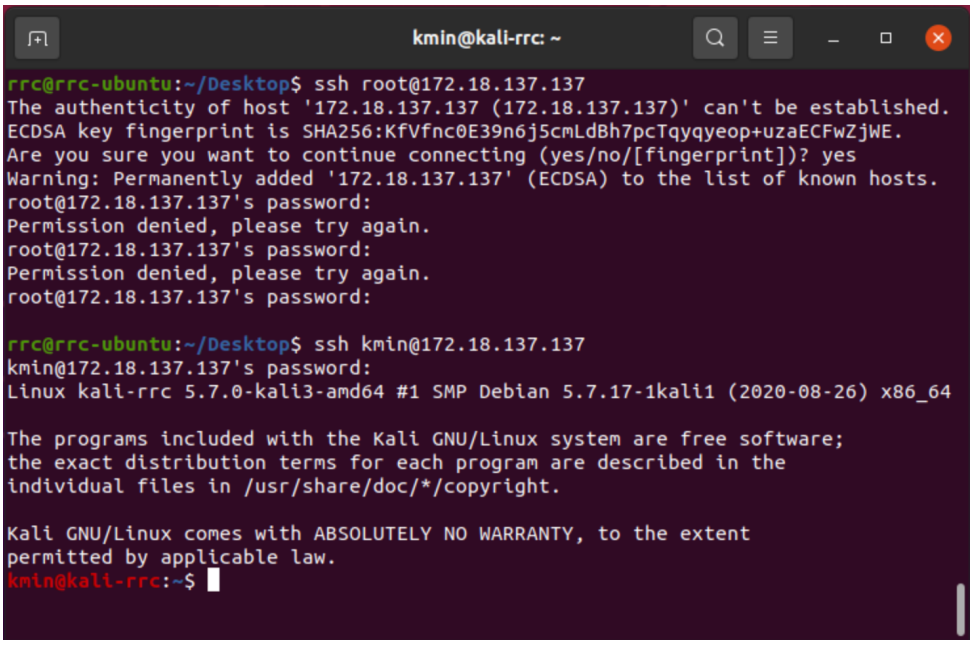
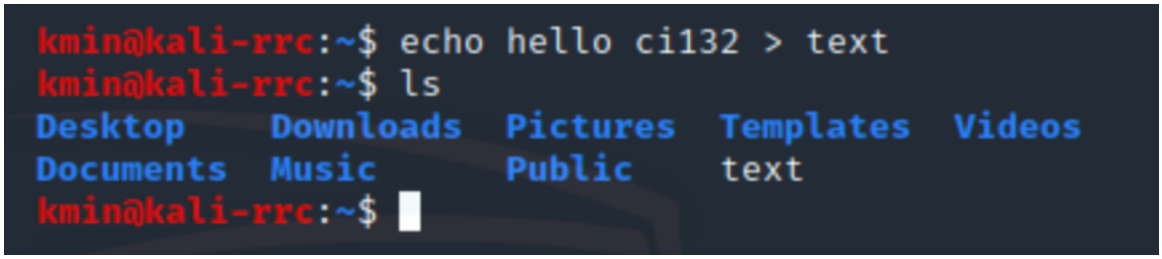
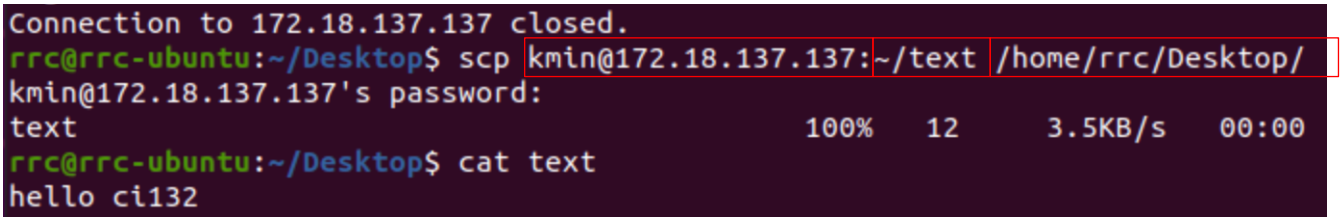
1. You can block a user from logging in with a password using ssh. You will just need to edit the “PermitRootLogin yes” to “PermitRootLogin no” in “/etc/ssh/sshd_config”
2. As we changed to no login with password, we may need to create a new user that we can ssh with a password, but the new password must be unique.



3. You can copy a file securely using “scp” command. You may need to create a file on the server side to try this out.



4. Use the “scp” command to copy from the server side. To receive file from the server can be written in “username@ipaddress:file_path_to_copy_from path_to_paste”



5. Now, you can send the file back to the server securely using the “scp” command. To send a file to the server, the format to write the command is “file_path_to_copy_from username@ipaddress:path_to_paste”

```
rrc@rrc-ubuntu:~/Desktop$ nano text
rrc@rrc-ubuntu:~/Desktop$ scp /home/rrc/Desktop/text kmin@172.18.137.137:~/Deskt
op/text2
kmin@172.18.137.137's password:
text                                          100%   26    22.6KB/s   00:00
kmin@kali-rrc:~$ cat ~/Desktop/text2
hello ci132
linux is fun
```

6. You can log in using ssh without a password since we changed the password login to a no. You may need to create a new account and generate new keys in your current user home directory using the “keygen” command.

7. After you generate your new keys, you may need to put those keys in a special location in order for the passwordless authentication to work, which is in the file path “/home/user/.ssh/authorized_keys”. So create a new file in your new user’s account.

```
kmin@kali-rrc:/home/bill$ mkdir .ssh
mkdir: cannot create directory '.ssh': Permission denied
kmin@kali-rrc:/home/bill$ sudo mkdir .ssh
kmin@kali-rrc:/home/bill$ ls -a
.   .bash_logout   .bashrc.original   .face.icon   .ssh
..  .bashrc        .face              .profile     .zshrc
kmin@kali-rrc:/home/bill$ ls -la
total 52
drwxr-xr-x 3 bill bill  4096 Feb 15 12:04 .
drwxr-xr-x 4 root root  4096 Feb 15 11:59 ..
-rw-r--r-- 1 bill bill   220 Feb 15 11:59 .bash_logout
-rw-r--r-- 1 bill bill  3391 Feb 15 11:59 .bashrc
-rw-r--r-- 1 bill bill  3526 Feb 15 11:59 .bashrc.original
-rw-r--r-- 1 bill bill 11759 Feb 15 11:59 .face
lrwxrwxrwx 1 bill bill     5 Feb 15 11:59 .face.icon → .face
-rw-r--r-- 1 bill bill   807 Feb 15 11:59 .profile
drwxr-xr-x 2 root root  4096 Feb 15 12:04 .ssh
-rw-r--r-- 1 bill bill  8238 Feb 15 11:59 .zshrc
```

8. Since we made the file, you may need to change file ownership using “chown” command. The command format is “sudo chown user:group filepath”

```
kmin@kali-rrc:/home/bill$ sudo chown bill:bill .ssh
kmin@kali-rrc:/home/bill$ ls -la
total 52
drwxr-xr-x 3 bill bill  4096 Feb 15 12:04 .
drwxr-xr-x 4 root root  4096 Feb 15 11:59 ..
-rw-r--r-- 1 bill bill   220 Feb 15 11:59 .bash_logout
-rw-r--r-- 1 bill bill  3391 Feb 15 11:59 .bashrc
-rw-r--r-- 1 bill bill  3526 Feb 15 11:59 .bashrc.original
-rw-r--r-- 1 bill bill 11759 Feb 15 11:59 .face
lrwxrwxrwx 1 bill bill     5 Feb 15 11:59 .face.icon → .face
-rw-r--r-- 1 bill bill   807 Feb 15 11:59 .profile
drwxr-xr-x 2 bill bill  4096 Feb 15 12:04 .ssh
-rw-r--r-- 1 bill bill  8238 Feb 15 11:59 .zshrc
```

9. You may also need to change the file permission in order for the file to work. Only the current owner should have access to it.

```
kmin@kali-rrc:/home/bill$ sudo chmod 700 .ssh
kmin@kali-rrc:/home/bill$ ls -la
total 52
drwxr-xr-x 3 bill bill  4096 Feb 15 12:04 .
drwxr-xr-x 4 root root  4096 Feb 15 11:59 ..
-rw-r--r-- 1 bill bill   220 Feb 15 11:59 .bash_logout
-rw-r--r-- 1 bill bill  3391 Feb 15 11:59 .bashrc
-rw-r--r-- 1 bill bill  3526 Feb 15 11:59 .bashrc.original
-rw-r--r-- 1 bill bill 11759 Feb 15 11:59 .face
lrwxrwxrwx 1 bill bill     5 Feb 15 11:59 .face.icon → .face
-rw-r--r-- 1 bill bill   807 Feb 15 11:59 .profile
drwx------ 2 bill bill  4096 Feb 15 12:04 .ssh
-rw-r--r-- 1 bill bill  8238 Feb 15 11:59 .zshrc
```

10. Now, transfer the newly generated keys to “.ssh” folder we just made for the new user. The file path you should paste is “.ssh/authorized_keys” which will create a new folder named “authorized_keys” in “.ssh” folder. Make sure you give ownership to the new user and the permission is “644” to that folder.

```
kmin@kali-rrc:/home/bill$ sudo chown bill:bill /home/bill/.ssh/authorized_k
eys
kmin@kali-rrc:/home/bill$ ls -l /home/bill/.ssh/
ls: cannot open directory '/home/bill/.ssh/': Permission denied
```

11. Now, the server is completely set up to use the passwordless authentication method. Copy your private key securely on the machine you want the client to be.

12. The command to use ssh without password may need an option “-i” and your password file path. Now, you can use the passwordless authentication method on your machine.

